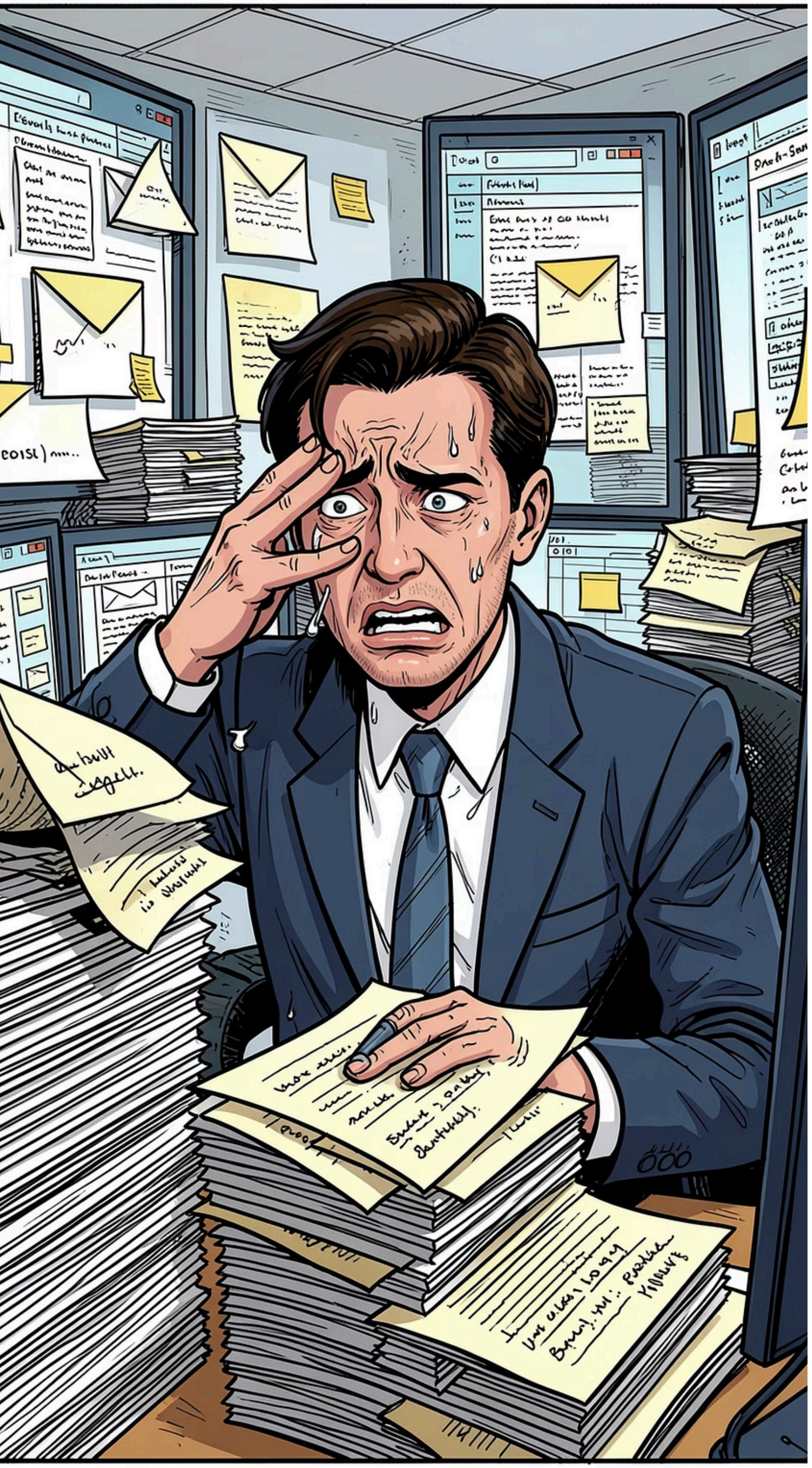
PEER REVIEW

AI Executive Assistant Agent

An AI agent that lives in your Gmail inbox



THE PROBLEM

Email management costs time — a lot of time

Every meeting request requires: reading, checking the calendar, drafting a reply, and following up. For a single email, that is **10–15 minutes** — multiplied by dozens of messages per week.

Read & Understand

Extract intent and scheduling request

Check calendar

Manually match availability

Draft response

Contextually appropriate, in the sender's language

Follow up

Manually create the appointment

THE SOLUTION

An AI agent in your inbox

The agent monitors your Gmail inbox, classifies every incoming emails and acts independently:

Read, Classify & extract

- Recognise intent, language and meeting request or meeting confirmation
- Extract informations like date and time

Check Calendar

Google Calendar API in real time

Generate answer and save Draft

Generate a context-aware reply as a Gmail draft — one click to send

Create Event

After sending by a human, the agent automatically adds the calendar entry

WHAT SETS US APART

Not script-based — but thinking

The agent uses a **ReAct loop** (Reason + Act) to reason step by step. If anything is unclear, it asks — instead of guessing or declining.

Our agent

ReAct loop with 9 steps, understands nuances, asks for clarification when needed

ARCHITECTURE

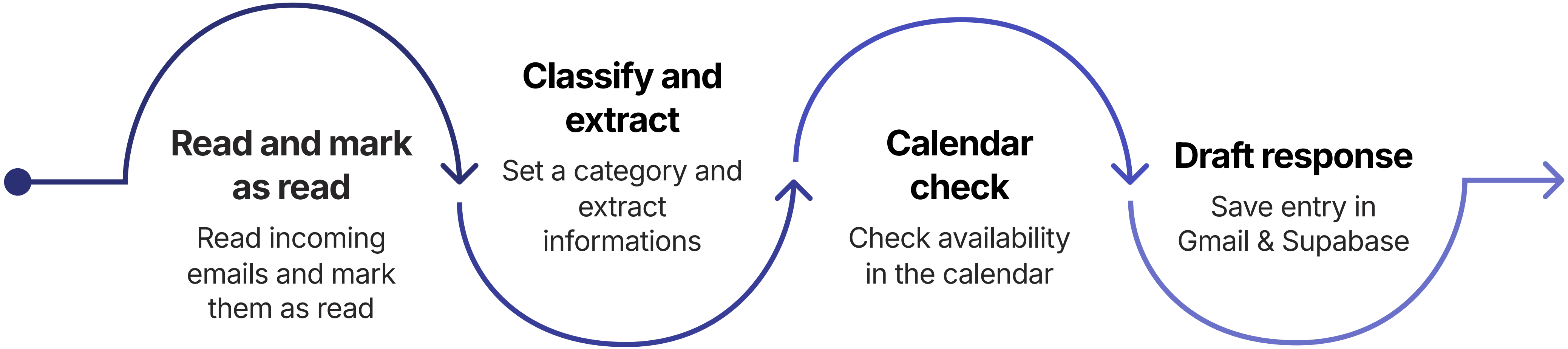
Technical Building Blocks

Pipeline Overview

The **AgentOrchestrator** controls a 9-step pipeline via the system prompt. Each tool is isolated and testable.

LIVE DEMO

Path: Specific meeting suggestion



After sending the draft by a human, the agent recognises the email in the **Sent** folder and automatically creates the calendar entry in Google Calendar — without any further action.

LIVE DEMO

Edge Cases: Busy & Vague

Scenario A — Slot taken

Email: “Can we meet Tuesday at 2pm?" — but 14:00 is already blocked.

→ The agent suggests **3 alternative slots**, evenly spread across the day, in the sender's language.

Scenario B — Vague request

Email: “Können wir uns nächste Woche am Nachmittag treffen??"

→ The agent replies in **German** and asks for the day and time precisely — without declining the meeting.

Live highlights: Automatic language detection · Supabase memory table · Draft as a reply thread in Gmail

KEY DECISIONS

Technical decisions

1

Language from email body

Each email is replied to in its own language. Use email body language instead of email subject language.

2

LLM for reasoning, Python for date parsing

LLM extracts raw strings (“next Friday", “14h30"). Parsing with `dateparser` in Python — testable and deterministic.

3

start_iso gates the draft path

Three-branch logic: confirmed / busy / vague — neatly covers all realistic cases.

REFLECTION

The Biggest Insight

> The real engineering challenge is not building the tools — it is writing a prompt that makes the agent's behaviour predictable and auditable.

LLM agents are inherently non-deterministic. Even with a detailed system prompt, the model may occasionally skip a step, use or change the order of tool calls.

REFLECTION RELIABILITY

How it has been improved?

Here are three key principles:

1. **A numbered, prescriptive prompt, not a descriptive one**

   Instead of describing what the agent can do, the prompt lists exactly what it must do and in what order

2. **Critical operations moved into the tool code**

   Some function calls are nested within others in Python. The LLM calls a single tool; the code performs the multiple operations.

3. **Checks in the tools, not in the prompt**

   The Python code itself validates certain conditions — rather than relying on the LLM.

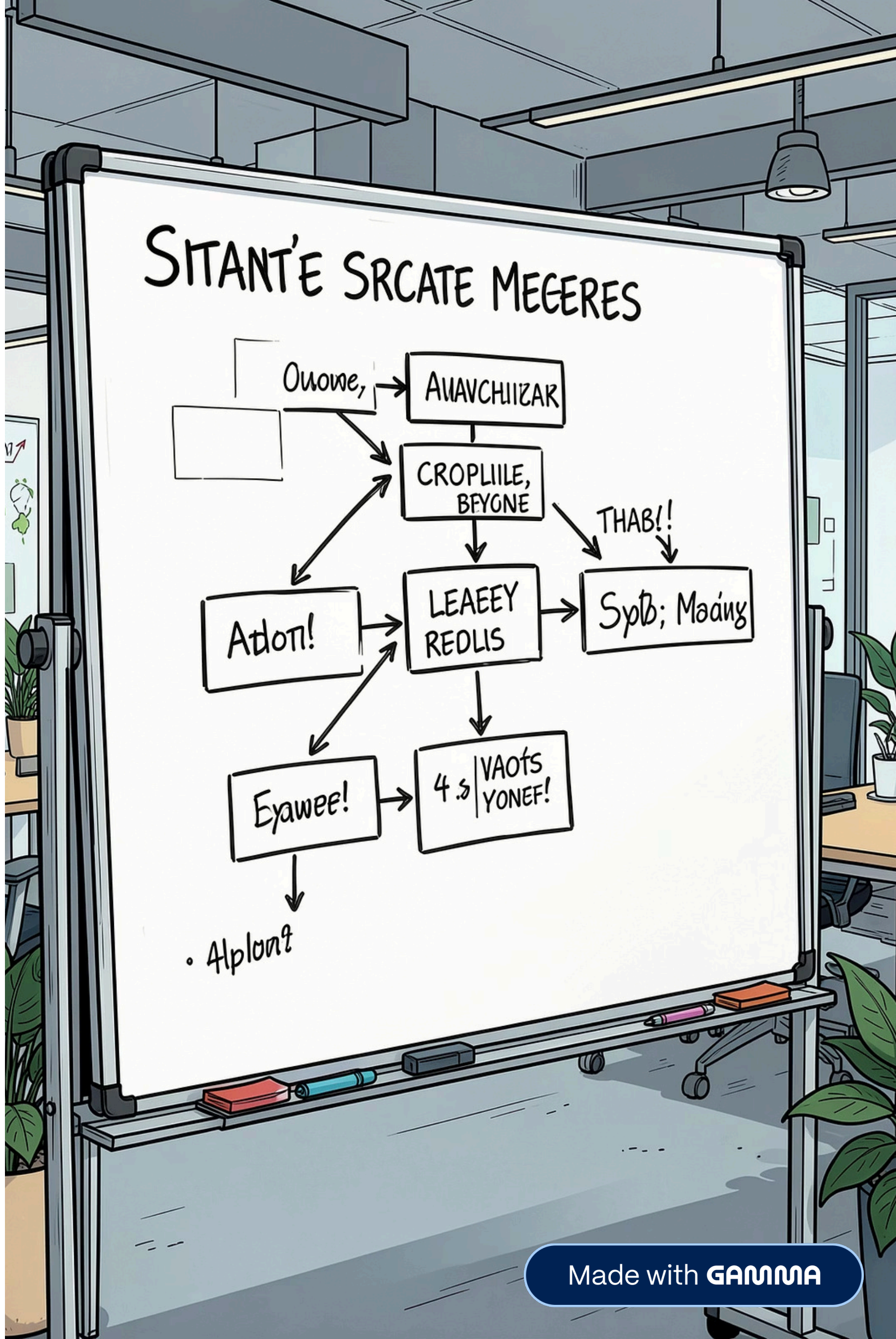
> **Making an LLM deterministic is impossible — but we can reduce the scope for ambiguity. The prompt specifies what to do and in what order. The tools guarantee how — the critical effects are in the code, not in the instructions.**

REFLECTION WHAT I WOULD DO DIFFERENTLY

Account for meeting cancellations

Instead of only handling meeting creation and confirmation, extend the system to detect cancellation intents

This would ensure that the calendar remains a single source of truth and stays fully synchronized with real user intent across the entire email lifecycle.



Thank you for listening!

Questions?